

Haris Riaz

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Java (8 yrs), C (9 yrs), Scala (2 yrs), Python (9 yrs), SQL (7 yrs)

EDUCATION

University of Arizona (CGPA: 3.91 /4.0) <i>PhD in Computer Science, Minor in Cognitive Science</i> <i>Advisor: Mihai Surdeanu</i> <i>Research Topic: Integrating Expert Knowledge into LLMs via Neurosymbolic Methods.</i>	Tucson, AZ Jan. 2022 – May 2026
University of Arizona (CGPA: 3.91/4.0) <i>MS in Computer Science</i>	Tucson, AZ Jan. 2022 – Dec 2024
National University of Sciences & Technology (CGPA: 3.75/4.0) <i>Bachelor of Science in Computer Science.</i> <i>Thesis: Handwritten Sequence Recognition with Time Series Transformers.</i>	Islamabad, PK Sep. 2017 – June 2021

EXPERIENCE

Applied Scientist Intern <i>Amazon Web Services</i> - Intern with Amazon Science Agentic AI team. - Worked on PRISM: Parametric Reasoning via Implicit Symbolic Modeling, a neurosymbolic reinforcement learning approach for verification and correction of an LLM's chain of thought. Evaluated on TravelPlanner and TauBench. Preparing Publication for ACL 2026 ARR.	May 2025 – Sep 2025 Seattle, WA
Applied Scientist Intern <i>Amazon Web Services</i> - Interned with Amazon Science Bedrock team - Focused on techniques for scaling Diverse Synthetic Data generation for Large Language Model (LLM) training. - Developed a Meta-prompting algorithm for creating synthetic data from scratch (using random seed keywords) & creating synthetic data by leveraging existing corpora with a RAG-Guided-Search approach. - Created a high quality, fully synthetic & formally diverse Instruction Tuning dataset of 25 million tokens. - Proposed and implemented metrics for measuring the formal diversity of LLM generated synthetic data. - This work resulted in a paper accepted at ACL 2025.	May 2024 – September 2024 Arlington, VA
Graduate Data Scientist Intern <i>Kaiser Permanente</i> - Focused on the development of efficient NLP models for identifying social health indicators from clinical text, aiding in comprehensive patient health assessments. - Built a diverse labeled training dataset of over 1 million provider notes using heuristic annotations, zero-shot learning with LLMs, human-in-the-loop active learning and multiple weak supervision strategies. - Used this dataset to train an NLP model that achieved over a 90% F1 score in expert evaluations, a state of the art in pinpointing key social determinants of health. - Developed and optimized high-performance distributed queries for the Epic Clarity database, enabling efficient retrieval and processing of billions of records for advanced machine learning analysis. - Gained hands-on experience with Hive, AzureML, Azure Data Factory & Azure Synapse Pipelines, Snorkel Flow/Snorkel AI, John Snow Labs' NLP and Docker deployment.	June 2023 – August 2023 San Francisco, CA
Graduate Research Assistant <i>University of Arizona</i> - Researched novel ways to understand people's opinions through text "Predicting What the Locals will Predict" (PWLP). This project is part of the DARPA Habitus initiative. - Worked on an unsupervised "co-ranking" method for identifying local domain experts based upon graphical representations of named entities, and random walk algorithms for ranking those entities.	May 2022 – June 2023 Tucson, AZ

PUBLICATIONS

MetaSynth: Meta-Prompting-Driven Agentic Scaffolds for Diverse Synthetic Data Generation. *In The Findings of the Association for Computational Linguistics (ACL 2025).*

ELLEN: Extremely Lightly Supervised Learning For Efficient Named Entity Recognition. *In The 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024).*

Best of Both Worlds: A Pliable and Generalizable Neuro-Symbolic Approach for Relation Classification. *In The 2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024 Findings).*

Deep neural network techniques in the calibration of space-charge distortion fluctuations for the ALICE TPC. *In The 25th International Conference on Computing in High-Energy Physics (CHEP 2021).*

PUBLICATIONS UNDER REVIEW

Say Less, Mean More: Leveraging Pragmatics in Retrieval-Augmented Generation. *In Review for ACL 2026.*

PRISM: Parametric Reasoning via Implicit Symbolic Modeling. *In Review for ACL 2026.*

PEER REVIEWING EXPERIENCE

- Reviewed **3 papers** for **The Second Workshop on Pattern-based Approaches to NLP in the Age of Deep Learning (Pan-DL), 2023.**
- Reviewed **1 paper** for the **2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2024** and **1 paper** for **The Thirteenth International Conference on Learning Representations (ICLR) 2025.**

AWARDS

Galileo Circle Scholarship Spring 2025
University of Arizona *Tucson, AZ*

- The Galileo Circle awards scholarships to undergraduate and graduate students who demonstrate exceptional potential in the physical, mathematical, environmental, cognitive, and life sciences.

Lightning talk @NeurIPS 2024 MusIML workshop December 2024
Neural Information Processing Systems (NeurIPS 2024) *Vancouver, BC*

- Our work on “Leveraging Pragmatics for Retrieval Augmented Generation” is chosen as a contributed lightning talk (oral presentation) at the MusIML workshop co-located with NeurIPS 2024.

College of Science Grad Fellowship Spring 2024
University of Arizona *Tucson, AZ*

Stanford Treehacks 2024, 2025 February 2024, 2025
Stanford University *Stanford, CA*

- Received a bursary covering all associated costs to participate in the premier U.S. collegiate hackathon, Stanford Treehacks in both 2024 and 2025.

AI Talent Bursary May 2022
Alberta Machine Intelligence Institute (AMII) *Edmonton, AB*

- Awarded \$1500 CAD for attending AI week organized by Alberta Machine Intelligence Institute (AMII).

Rector’s Gold Medal shortlist June 2021
National University of Sciences & Technology *Islamabad, PK*

- Among 3 people shortlisted for Rector’s Gold Medal for best final year CS project.

Turkish Aerospace Industries (TAI) Summer Program July 2020
Turkish Aerospace Academy *Ankara, Turkey*

- Nominated for a fully funded vocational summer program at Turkish Aerospace Academy based upon demonstrated aptitude in technical disciplines.

SKILLS

Java • Python • C • Scala • SQL • JavaScript • Algorithms • PyTorch • Machine Learning • Convolutional Neural Networks • Transformers • NLP • Image Processing • Time Series • OpenCV • Git • HTML/CSS • D3.js • Maven